## DK1 imbalance: methods

github.com/jacobazu/dk1-imbalance-forecasting · October 2026

Forecasts of the chance that DK1 is regulated up, down or not at all in each quarter-hour, and of the spread between the imbalance price and spot. Every three hours for the next four hours, and once a day for tomorrow.

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| Inputs | Calendar, the spot price, wind and solar forecasts made 48 hours ahead, recent imbalances, and real-time grid data for the next hour. Before the auction the price forecast stands in for spot. |
| Models | Multinomial lasso (Tibshirani, 1996; Friedman et al., 2010), XGBoost (Chen and Guestrin, 2016) and their average, refitted weekly. Benchmark: how often each direction happened over the last 28 days. Spread: past spreads at a similar spot price, weighted by the direction probabilities. |
| Scoring | Log loss and pinball loss against the benchmark, Diebold–Mariano tests with a Newey–West variance (Diebold and Mariano, 1995; Newey and West, 1987), calibration, and the value of picking a side (Bottieau et al., 2020). |
| Day-ahead to imbalance | Long or short 1 MW per quarter on the day-ahead auction, settled at the imbalance price, after fees. Compared with always long, always short and the benchmark with Newey–West and Holm. Paper trading. |
| Rules | Each forecast only uses data published when it is made, and the backtest uses the same inputs. Test rules are written before the test. |

## References


Bottieau, J. et al. (2020). Very-short-term probabilistic forecasting for a risk-aware participation in the single price imbalance settlement. *IEEE Transactions on Power Systems*.

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